



```

name: SimulationBaltic
log: C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
log type: smcl
opened on: 26 Aug 2026, 15:05:52

1 .
2 . asdoc, text(\par \qc Financial Sustainability of the Care for the Disabled: A Simula
> tion of Benefits for Older People in the Baltic States up to 2036 with data from SHA
> RE and Eurostat) fs(12) save(report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

3 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

4 .
5 . *>>>PREPROCESSING
6 .
7 . foreach num of numlist 8(1)9{
2. clear all
3. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
> 7\sharew`num'_rel9-0-0_ALL_datasets_stata\"
4. use "sharew`num'_rel9-0-0_gv_imputations.dta"
5. keep if implicat==1
6. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_health.dta"
7. isid mergeid
8. drop _merge
9. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_weights.dta"
10. isid mergeid
11. drop _merge
12.
8 . if `num'==8{
13. gen Qyear=2020
14. }
15. if `num'==9{
16. gen Qyear=2022
17. }
18.
9 . keep if country==57 | country==48 | country==35
19.
10. save "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
> \W`num'_final.dta", replace
20. }
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew8_rel9-0-
> 0 ALL_datasets_stata
(214,780 observations deleted)
(label orienti already defined)
(label eurod already defined)
(label numeracy already defined)
(label gali already defined)
(label language already defined)
(label country already defined)

Result                                Number of obs
-----
Not matched                           0
Matched                               53,695   (_merge==3)

(label country already defined)
(label language already defined)

```

Result	Number of obs	
Not matched	0	
Matched	53,695	(_merge==3)

(47,632 observations deleted)

file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
7\W8_final.dta saved

C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew9_rel9-0-

> 0 ALL datasets stata

(277,788 observations deleted)

(label **orienti** already defined)

(label **eurod** already defined)

(label **numeracy** already defined)

(label **gali** already defined)

(label **language** already defined)

(label **country** already defined)

Result	Number of obs	
Not matched	0	
Matched	69,447	(_merge==3)

(label **country** already defined)

(label **language** already defined)

Result	Number of obs	
Not matched	0	
Matched	69,447	(_merge==3)

(61,831 observations deleted)

file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
7\W9_final.dta saved

11.

12. clear all

13. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7"

14. use "W8_final.dta"

15. merge 1:1 mergeid using "weightslong.dta", keep(match)
(label **country** already defined)

Result	Number of obs	
Not matched	0	
Matched	5,101	(_merge==3)

16. drop _merge

17. save "W8_final.dta", replace
file W8_final.dta saved

18.

19. clear all

```
20. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
21. use "W9_final.dta"
```

```
22. merge 1:1 mergeid using "weightslong.dta", keep(match)
    (label country already defined)
```

Result	Number of obs	
Not matched	0	
Matched	4,921	(_merge ==3)

```
23. drop _merge
```

```
24. save "W9_final.dta", replace
    file W9_final.dta saved
```

```
25.
```

```
26. clear all
```

```
27. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
28. use "W8_final.dta"
```

```
29. append using "W9_final.dta"
    (variable psu was str10, now str11 to accommodate using data's values)
    (variable ssu was str6, now str13 to accommodate using data's values)
    (label pstat already defined)
    (label language already defined)
    (label country already defined)
    (label Hrespt already defined)
    (label pcskill already defined)
    (label IMPflag already defined)
    (label orienti already defined)
    (label bfi10 already defined)
    (label thospital already defined)
    (label satisfied already defined)
    (label often already defined)
    (label yedu already defined)
    (label fdistress already defined)
    (label writing already defined)
    (label reading already defined)
    (label hearing already defined)
    (label eyesight already defined)
    (label mstat already defined)
    (label sphus already defined)
    (label noyes already defined)
    (label hnrsc already defined)
    (label undersq already defined)
    (label clarif already defined)
    (label willans already defined)
    (label otrf already defined)
    (label empstat already defined)
    (label cjs already defined)
    (label isced already defined)
    (label gender already defined)
    (label num already defined)
    (label gali already defined)
    (label meat already defined)
    (label legeggs already defined)
    (label fruit already defined)
    (label numeracy already defined)
    (label numeracy2 already defined)
    (label eurod already defined)
    (label dairyp already defined)
    (label weekday already defined)
    (label specday already defined)
    (label dkrf already defined)
    (label dummi already defined)
```

```

(label zero_one already defined)
(label loneli already defined)
(label chronic2 already defined)
(label sphus2 already defined)
(label bmi2 already defined)
(label mobilit2 already defined)
(label mobilit3 already defined)
(label adl2 already defined)
(label iadl2 already defined)
(label phactiv already defined)

30.
31. save "dataset.dta", replace
    file dataset.dta saved

32.
33. clear all

34.
35. use "dataset.dta"

36.
37. *>>>PREPARATION
38.
39. generate income = thinc-ypen3

40.
41. egen id= group(mergeid)

42.
43. keep ypen3 gender age country income gali id Qyear

44.
45. replace income=0 if income<0
    (44 real changes made)

46.
47. drop if ypen3>7000
    (7 observations deleted)

48.
49. *additional age categorization to match Eurostat categories
50. generate ageD=0

51. replace ageD=1 if age>=55 & age<=64
    (2,793 real changes made)

52. replace ageD=2 if age>=65 & age<=74
    (3,142 real changes made)

53. replace ageD=3 if age>=75 & age<=84
    (2,554 real changes made)

54. replace ageD=4 if age>=85
    (876 real changes made)

55. label define ageDl 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Bet
    > ween 65 and 74 years old" 3 "Between 75 and 84 years old" 4 "Older than 85 years"

56. label values ageD ageDl

```

```

57.
58. generate ageE=0

59. replace ageE=1 if age>=55 & age<=64
   (2,793 real changes made)

60. replace ageE=2 if age>=65
   (6,572 real changes made)

61. label define agel 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Olde
   > r than 64 years"

62. label values ageE agel

63.
64. *>>>DESCRIPTIVES
65.
66. tabulate country

```

Country identifier	Freq.	Percent	Cum.
Estonia	5,573	55.65	55.65
Lithuania	2,429	24.25	79.90
Latvia	2,013	20.10	100.00
Total	10,015	100.00	

```

67.
68. asdoc codebook ypen3 gender age country income gali id Qyear, compact save(report.do
   > c) append

```

Variable	Obs	Unique	Mean	Min	Max	Label
ypen3	10015	382	232.8375	0	6816	Disability pensions/benefits
gender	10015	2	1.632052	1	2	Gender
age	10015	65	69.76246	34	102	Age
country	10015	3	42.57494	35	57	Country identifier
income	10015	4468	10071.76	0	170698	
gali	10015	2	.5899151	0	1	Limitation with activities
id	10015	5101	2556.266	1	5101	group(mergeid)
Qyear	10015	2	2020.982	2020	2022	

Click to Open File: [report.doc](#)

```

69.
70. asdoc codebook, save(report.doc) append

```

country	Country identifier
Type: Numeric (byte)	
Label: country	
Range: [35,57]	
Unique values: 3	
Units: 1	
Missing .: 0/10,015	
Tabulation: Freq.	Numeric Label
5,573	35 Estonia
2,429	48 Lithuania
2,013	57 Latvia
ypen3	Disability pensions/benefits

Type: Numeric (**double**)

Range: [0,6816] Units: 1.000e-07
 Unique values: 382 Missing .: 0/10,015

Mean: 232.838
 Std. dev.: 885.214

Percentiles: 10% 25% 50% 75% 90%
 0 0 0 0 276

gender **Gender**

Type: Numeric (**byte**)
 Label: **gender**

Range: [1,2] Units: 1
 Unique values: 2 Missing .: 0/10,015

Tabulation: Freq. Numeric Label
 3,685 1 Male
 6,330 2 Female

age **Age**

Type: Numeric (**int**)
 Label: **num**, but 65 nonmissing values are not labeled

Range: [34,102] Units: 1
 Unique values: 65 Missing .: 0/10,015

Examples: 60
 66
 73
 80

gali **Limitation with activities**

Type: Numeric (**byte**)
 Label: **gali**

Range: [0,1] Units: 1
 Unique values: 2 Missing .: 0/10,015

Tabulation: Freq. Numeric Label
 4,107 0 Not limited
 5,908 1 Limited

Qyear **(unlabeled)**

Type: Numeric (**float**)

Range: [2020,2022] Units: 1
 Unique values: 2 Missing .: 0/10,015

Tabulation: Freq. Value
 5,100 2020
 4,915 2022

income **(unlabeled)**

Type: Numeric (**float**)

Range: [0,170697.95] Units: 1.000e-20
Unique values: 4,468 Missing .: 0/10,015

Mean: 10071.8
Std. dev.: 8975.29

Percentiles: 10% 25% 50% 75% 90%
 1968 4900 7750 12860 19971.8

id **group (mergeid)**

Type: Numeric (**float**)

Range: [1,5101] Units: 1
Unique values: 5,101 Missing .: 0/10,015

Mean: 2556.27
Std. dev.: 1474.24

Percentiles: 10% 25% 50% 75% 90%
 512 1278 2559 3835 4595

ageD **(unlabeled)**

Type: Numeric (**float**)
Label: **ageD1**

Range: [0,4] Units: 1
Unique values: 5 Missing .: 0/10,015

Tabulation:	Freq.	Numeric	Label
	650	0	Less than 55 years old
	2,793	1	Between 55 and 64 years old
	3,142	2	Between 65 and 74 years old
	2,554	3	Between 75 and 84 years old
	876	4	Older than 85 years

ageE **(unlabeled)**

Type: Numeric (**float**)
Label: **age1**

Range: [0,2] Units: 1
Unique values: 3 Missing .: 0/10,015

Tabulation:	Freq.	Numeric	Label
	650	0	Less than 55 years old
	2,793	1	Between 55 and 64 years old
	6,572	2	Older than 64 years

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```

72. hist ypen3 if ypen3>0, scheme(s2color) xlabel(0(1000)7000,angle(0) labsize(small) gr
> id) ylabel(0.000(0.0002)0.001,angle(0) labsize(small) grid format(%9.0gc))
(bin=30, start=40, width=225.86667)

73. graph export benefits.svg, replace height(2400)
(file benefits.svg not found)
file benefits.svg saved as SVG format

74. vioplot age, over(gender) over(country) scheme(s2color) ytitle("Number of respondent
> s", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)

75. graph export sample.svg, replace height(2400)
(file sample.svg not found)
file sample.svg saved as SVG format

76. vioplot age if ypen3>0, over(gender) over(country) scheme(s2color) ytitle("Number of
> respondents", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)

77. graph export sampleDis.svg, replace height(2400)
(file sampleDis.svg not found)
file sampleDis.svg saved as SVG format

78.

79. asdoc spearman ypen3 gender age country income gali id Qyear, save(report.doc) appen
> d

```

Number of observations = 10,015

	ypen3	gender	age	country	income	gali	id	Qyear
ypen3	1.0000							
gender	-0.0184	1.0000						
age	-0.1637	0.0735	1.0000					
country	-0.0932	-0.0113	-0.0870	1.0000				
income	-0.1245	-0.1481	-0.0799	-0.3425	1.0000			
gali	0.2259	0.0210	0.2934	-0.0195	-0.1472	1.0000		
id	-0.0918	-0.0058	-0.0739	0.8974	-0.2985	-0.0242	1.0000	
Qyear	-0.0373	0.0143	0.0877	0.0042	0.0780	0.0266	0.0037	1.0000

>

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```

80.
81. foreach ageC in ageD ageE{
2.     foreach num of numlist 35 48 57{
3.         foreach num2 of numlist 1 2{
4.             foreach num3 of numlist 0 1{
5.                 asdoc, text(\par for `ageC': Country `num
> ' , gender `num2' , limited `num3')
6.                 asdoc tabstat income if country==`num' &
> gender==`num2' & gali==`num3', by(`ageC') stat(mean), save(report.doc) append
7.             }
8.         }
9.     }
10. }
(File report.doc already exists, option append was assumed)
Click to Open File: report.doc
ageD

```

	income
Between_55~d	20571.48
Between_65~d	17941.57
Between_75~d	16346.79
Less_than_~d	12735.02
Older_than~s	11303.48

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ageD

	income
Between_55~d	12357.32
Between_65~d	13166.79
Between_75~d	13350.15
Less_than_~d	11962.99
Older_than~s	10109.23

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(File report.doc already exists, option **append** was assumed)

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aged

	income
Between_55~d	20010.27
Between_65~d	16784.18
Between_75~d	13343.93
Less_than_~d	9293.497
Older_than~s	7288.414

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aged

	income
Between_55~d	13631.43
Between_65~d	10804.44
Between_75~d	10525.3
Less_than_~d	8400.997
Older_than~s	7017.735

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aged

	income
Between_55~d	7846.787
Between_65~d	11125
Between_75~d	9710.709
Less_than_~d	9511.654
Older_than~s	7384.786

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aged

	income
Between_55~d	9433.906
Between_65~d	7559.58
Between_75~d	7553.142
Less_than_~d	7746.841
Older_than~s	6844.66

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aged

	income
Between_55~d	7358.451
Between_65~d	9387.978
Between_75~d	8545.833
Less_than_~d	6252.98
Older_than~s	5745.49

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aged

	income
Between_55~d	9299.859
Between_65~d	7512.068
Between_75~d	6602.537
Less_than_~d	6118.597
Older_than~s	4264.137

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aged

	income
Between_55~d	3934.908
Between_65~d	7333.089
Between_75~d	9284.544
Less_than_~d	9234.106
Older_than~s	7375.984

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aged

	income
Between_55~d	5057.722
Between_65~d	5015.384
Between_75~d	8295.423
Less_than_~d	7732.316
Older_than~s	7267.691

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aged

	income
Between_55~d	7083.539
Between_65~d	6286.68
Between_75~d	8461.983
Less_than_~d	6394.999
Older_than~s	5576.932

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aged

	income
Between_55~d	5664.375
Between_65~d	5343.996
Between_75~d	6306.117
Less_than_~d	5626.213
Older_than~s	4190.057

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ageE

	income
Between_55~d	20571.48
Less_than_~d	17941.57
Older_than~s	14906.06

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ageE

	income
Between_55~d	12357.32
Less_than_~d	13166.79
Older_than~s	12353.23

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Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	20010.27
Less_than_~d	16784.18
Older_than~s	11453.5

Click to Open File: [report.doc](#)
 (File report.doc already exists, option **append** was assumed)
Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	13631.43
Less_than_~d	10804.44
Older_than~s	8946.796

Click to Open File: [report.doc](#)
 (File report.doc already exists, option **append** was assumed)
Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	7846.787
Less_than_~d	11125
Older_than~s	9552.095

Click to Open File: [report.doc](#)
 (File report.doc already exists, option **append** was assumed)
Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	9433.906
Less_than_~d	7559.58
Older_than~s	7541.938

Click to Open File: [report.doc](#)
 (File report.doc already exists, option **append** was assumed)
Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	7358.451
Less_than_~d	9387.978
Older_than~s	7588.958

Click to Open File: [report.doc](#)
 (File report.doc already exists, option **append** was assumed)
Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	9299.859
Less_than_~d	7512.068
Older_than~s	5866.109

Click to Open File: [report.doc](#)
 (File report.doc already exists, option **append** was assumed)
Click to Open File: [report.doc](#)
 ageE

	income
Between_55~d	3934.908
Less_than_~d	7333.089
Older_than~s	9193.949

Click to Open File: [report.doc](#)

(File report.doc already exists, option **append** was assumed)

Click to Open File: [report.doc](#)

ageE

	income
Between_55~d	5057.722
Less_than_~d	5015.384
Older_than~s	7950.08

Click to Open File: [report.doc](#)

(File report.doc already exists, option **append** was assumed)

Click to Open File: [report.doc](#)

ageE

	income
Between_55~d	7083.539
Less_than_~d	6286.68
Older_than~s	7731.15

Click to Open File: [report.doc](#)

(File report.doc already exists, option **append** was assumed)

Click to Open File: [report.doc](#)

ageE

	income
Between_55~d	5664.375
Less_than_~d	5343.996
Older_than~s	5703.856

Click to Open File: [report.doc](#)

82.

83. *>>>ANALYSIS

84.

85. xtset id Qyear

Panel variable: **id** (unbalanced)

Time variable: **Qyear**, 2020 to 2022, but with gaps

Delta: **1 unit**

86.

87. *for disability

88. xttobit ypen3 i.gender i.ageD i.gali income i.country, ll(0) iterate(25) tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0: Log likelihood = **-21562.521**
 Iteration 1: Log likelihood = **-13222.906**
 Iteration 2: Log likelihood = **-13170.91**
 Iteration 3: Log likelihood = **-13170.391**
 Iteration 4: Log likelihood = **-13170.391**

Fitting full model:

Iteration 0: Log likelihood = **-21003.9**
 Iteration 1: Log likelihood = **-12648.454**
 Iteration 2: Log likelihood = **-12233.36**
 Iteration 3: Log likelihood = **-12215.992**
 Iteration 4: Log likelihood = **-12215.928**
 Iteration 5: Log likelihood = **-12215.928**

Obtaining starting values for full model:

Iteration 0: Log likelihood = **-80432.953**
 Iteration 1: Log likelihood = **-80374.31**
 Iteration 2: Log likelihood = **-80373.876**
 Iteration 3: Log likelihood = **-80373.876**

Fitting full model:

Iteration 0: Log likelihood = **-19495.179**
 Iteration 1: Log likelihood = **-14709.844** (not concave)
 Iteration 2: Log likelihood = **-12911.337**
 Iteration 3: Log likelihood = **-12042.141**
 Iteration 4: Log likelihood = **-11985.302**
 Iteration 5: Log likelihood = **-11984.256**
 Iteration 6: Log likelihood = **-11984.255**
 Iteration 7: Log likelihood = **-11984.255**

Random-effects tobit regression

Limits: Lower = **0**
 Upper = **+inf**

Group variable: **id**
 Random effects $u_i \sim \text{Gaussian}$

Number of obs = **10,015**
 Uncensored = **1,120**
 Left-censored = **8,895**
 Right-censored = **0**

Number of groups = **5,101**
 Obs per group:
 min = **1**
 avg = **2.0**
 max = **2**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-11984.255**

Wald chi2(9) = **810.68**
 Prob > chi2 = **0.0000**

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i	
> nterval]							
	gender						
	Female	-287.643	135.2631	-2.13	0.033	-552.7537	-
> 22.53216							
	ageD						
Between 55 and 64 years old		674.5725	220.543	3.06	0.002	242.3162	
> 1106.829							
Between 65 and 74 years old		-2665.123	249.0781	-10.70	0.000	-3153.307	-
> 2176.938							
Between 75 and 84 years old		-3295.505	260.9746	-12.63	0.000	-3807.006	-
> 2784.004							
Older than 85 years		-3137.637	306.8433	-10.23	0.000	-3739.039	-
> 2536.236							
	gali						
	Limited	3067.465	164.0657	18.70	0.000	2745.903	
> 3389.028							
	income	-.0813264	.0080558	-10.10	0.000	-.0971155	-
> .0655373							
	country						
	Lithuania	-1361.48	169.4983	-8.03	0.000	-1693.69	-
> 1029.269							
	Latvia	-2580.163	208.8472	-12.35	0.000	-2989.496	
> -2170.83							
	_cons	-2954.945	287.3196	-10.28	0.000	-3518.082	-
> 2391.809							
	/sigma_u	2498.727	91.03549	27.45	0.000	2320.301	
> 2677.153							
	/sigma_e	1874.845	58.8259	31.87	0.000	1759.548	
> 1990.142							

	rho			
> .6814528	.6398034	.0217323		.5964239

LR test of sigma_u=0: chibar2(01) = 463.35 Prob >= chibar2 = 0.000

89. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
 > title(DisabilityTobit) replace
 results.xls
 dir : seeout

90.

91. xtpoisson ypen3 i.gender i.ageD i.gali income i.country if ypen3>0, re vce(robust)
 note: you are responsible for interpretation of non-count dep. variable

Fitting Poisson model:

Iteration 0: Log pseudolikelihood = -510758.37
 Iteration 1: Log pseudolikelihood = -509695.99
 Iteration 2: Log pseudolikelihood = -509693.04
 Iteration 3: Log pseudolikelihood = -509693.04

Fitting full model:

Iteration 0: Log pseudolikelihood = -90029.433
 Iteration 1: Log pseudolikelihood = -81819.234
 Iteration 2: Log pseudolikelihood = -81485.43
 Iteration 3: Log pseudolikelihood = -81464.412
 Iteration 4: Log pseudolikelihood = -81464.282
 Iteration 5: Log pseudolikelihood = -81464.282

Random-effects Poisson regression
 Group variable: **id**

Number of obs = 1,120
 Number of groups = 769

Random effects u_i ~ Gamma

Obs per group:
 min = 1
 avg = 1.5
 max = 2

Log pseudolikelihood = -81464.282

Wald chi2(9) = 73075.05
 Prob > chi2 = 0.0000

(Std. err. adjusted for clusterin

> g on id)

	ypen3	Coefficient	Robust std. err.	z	P> z	[95% conf. i
> nterval]						
	gender					
	Female	-.1144357	.0626614	-1.83	0.068	-.2372498
> .0083784						
	ageD					
Between 55 and 64 years old		.3505128	.1000438	3.50	0.000	.1544306
> .5465951						
Between 65 and 74 years old		-.0381719	.2200633	-0.17	0.862	-.469488
> .3931442						
Between 75 and 84 years old		-.2965865	.2460516	-1.21	0.228	-.7788388
> .1856659						
Older than 85 years		-.1641732	.2578095	-0.64	0.524	-.6694706
> .3411241						
	gali					
	Limited	.0414885	.0798963	0.52	0.604	-.1151052
> .1980823						
	income	-3.51e-06	4.86e-06	-0.72	0.470	-.000013

> 6.01e-06

	country					
> .5385474	Lithuania	.3884312	.0765913	5.07	0.000	.238315
> .2666026	Latvia	.1006352	.0846788	1.19	0.235	-.0653322
> 7.59565	_cons	7.333101	.1339558	54.74	0.000	7.070553
> .0448174	/lnalpha	-.2402887	.0997321			-.4357599
> .9561721	alpha	.7864008	.0784294			.646773

LR test of alpha=0: chibar2(01) = 8.6e+05 Prob >= chibar2 = 0.000

92. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
 > title(DisabilityPoisson) append
results.xls
 dir : seeout

93. outreg2 using resultsNoStar.xls, excel dec(10) ctitle(DisabilityPoisson) noaster nos
 > e replace
resultsNoStar.xls
 dir : seeout

94.

95. *for income

96. xttoibit ypen3 i.gender i.ageE i.gali income i.country, ll(0) iterate(25) tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0: Log likelihood = -21562.521
 Iteration 1: Log likelihood = -13222.906
 Iteration 2: Log likelihood = -13170.91
 Iteration 3: Log likelihood = -13170.391
 Iteration 4: Log likelihood = -13170.391

Fitting full model:

Iteration 0: Log likelihood = -21016.972
 Iteration 1: Log likelihood = -12656.011
 Iteration 2: Log likelihood = -12239.864
 Iteration 3: Log likelihood = -12222.092
 Iteration 4: Log likelihood = -12222.03
 Iteration 5: Log likelihood = -12222.03

Obtaining starting values for full model:

Iteration 0: Log likelihood = -80443.279
 Iteration 1: Log likelihood = -80388.043
 Iteration 2: Log likelihood = -80387.672
 Iteration 3: Log likelihood = -80387.672

Fitting full model:

```

Iteration 0: Log likelihood = -19496.048
Iteration 1: Log likelihood = -14692.5 (not concave)
Iteration 2: Log likelihood = -13545.246
Iteration 3: Log likelihood = -12045.587
Iteration 4: Log likelihood = -11999.188
Iteration 5: Log likelihood = -11990.142
Iteration 6: Log likelihood = -11990.072
Iteration 7: Log likelihood = -11990.072
Iteration 8: Log likelihood = -11990.072

```

Random-effects tobit regression

Limits: Lower = 0
Upper = +inf

Group variable: **id**
Random effects $u_i \sim \text{Gaussian}$

Number of obs = 10,015
Uncensored = 1,120
Left-censored = 8,895
Right-censored = 0

Number of groups = 5,101
Obs per group:
min = 1
avg = 2.0
max = 2

Integration method: **mvaghermite**

Integration pts. = 12

Log likelihood = -11990.072

Wald chi2(7) = 797.62
Prob > chi2 = 0.0000

	ypen3	Coefficient	Std. err.	z	P> z	[95% conf. i	
> nterval]							
	gender						
> 44.22447	Female	-309.286	135.2379	-2.29	0.022	-574.3475	-
	ageE						
Between 55 and 64 years old		660.9327	220.8577	2.99	0.003	228.0596	
> 1093.806	Older than 64 years	-2970.574	235.0225	-12.64	0.000	-3431.209	-
> 2509.938							
	gali						
> 3370.135	Limited	3047.965	164.3756	18.54	0.000	2725.795	
> .0642995	income	-.0801059	.0080646	-9.93	0.000	-.0959123	-
	country						
> 1017.893	Lithuania	-1350.161	169.5277	-7.96	0.000	-1682.43	-
> 2139.595	Latvia	-2548.195	208.4733	-12.22	0.000	-2956.795	-
	_cons						
> 2394.483		-2958.879	287.9623	-10.28	0.000	-3523.274	-
	/sigma_u						
> 2682.524		2504.145	91.0116	27.51	0.000	2325.766	
> 1995.871	/sigma_e	1880.215	59.0092	31.86	0.000	1764.56	
	rho						
> .6810337		.6394832	.0216781			.5962159	

LR test of $\sigma_u=0$: chibar2(01) = 463.92

Prob >= chibar2 = 0.000


```
97. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(IncomeTobit) append
results.xls
dir : seeout
```

98.

```
99. xtpoisson ypen3 i.gender i.ageE i.gali income i.country if ypen3>0, re vce(robust)
note: you are responsible for interpretation of non-count dep. variable
```

Fitting Poisson model:

```
Iteration 0: Log pseudolikelihood = -535702.37
Iteration 1: Log pseudolikelihood = -535546.08
Iteration 2: Log pseudolikelihood = -535546.04
```

Fitting full model:

```
Iteration 0: Log pseudolikelihood = -95899.544
Iteration 1: Log pseudolikelihood = -82803.035
Iteration 2: Log pseudolikelihood = -81530.966
Iteration 3: Log pseudolikelihood = -81502.661
Iteration 4: Log pseudolikelihood = -81502.309
Iteration 5: Log pseudolikelihood = -81502.309
```

Random-effects Poisson regression
Group variable: **id**

Number of obs = **1,120**
Number of groups = **769**

Random effects u_i ~ Gamma

Obs per group:
min = **1**
avg = **1.5**
max = **2**

Log pseudolikelihood = **-81502.309**

Wald chi2(7) = **74617.49**
Prob > chi2 = **0.0000**

(Std. err. adjusted for clusterin

> g on **id**)

	ypen3	Coefficient	Robust std. err.	z	P> z	[95% conf. i
> nterval]						
	gender					
> .0059222	Female	-.1189755	.0637245	-1.87	0.062	-.2438732
	ageE					
Between 55 and 64 years old		.3505488	.1000543	3.50	0.000	.1544461
> .5466516	Older than 64 years	-.0382013	.2202289	-0.17	0.862	-.469842
> .3934394						
	gali					
> .1980949	Limited	.0414862	.0799038	0.52	0.604	-.1151224
> 6.02e-06	income	-3.51e-06	4.86e-06	-0.72	0.471	-.000013
	country					
> .519685	Lithuania	.3722408	.0752281	4.95	0.000	.2247965
> .2775911	Latvia	.11164	.0846705	1.32	0.187	-.054311
> 7.58033	_cons	7.318747	.1334633	54.84	0.000	7.057164
	/lnalpha	-.176007	.0927892			-.3578705
> .0058565						

	alpha	.8386121	.0778141	.6991636
> 1.005874				

LR test of alpha=0: chibar2(01) = 9.1e+05 Prob >= chibar2 = 0.000

100 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
 > title(IncomePoisson) append
results.xls
 dir : seeout

101 outreg2 using resultsNoStar.xls, excel dec(10) ctitle(IncomePoisson) noaster nose ap
 > pend
resultsNoStar.xls
 dir : seeout

102

103 log close SimulationBaltic
 name: **SimulationBaltic**
 log: C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
 > tput.smcl
 log type: **smcl**
 closed on: **26 Aug 2026, 15:06:52**
